Manish Kumar

Glenview, IL | Email | LinkedIn | Portfolio

SUMMARY

Senior Frontend Engineer with nearly 8 years of experience building production Angular, React, and TypeScript applications, enterprise design systems, and accessible (WCAG) component libraries. Migrated a legacy AngularJS application to modern Angular at Dun & Bradstreet, and built a shared UI platform at Medline adopted across 7+ applications and multiple teams. Recently designed and shipped a full-stack, AI-powered SaaS platform independently, from frontend architecture through payments and cloud deployment. Published researcher on component reusability, resilient frontend architecture, and AI-assisted development, currently pursuing a Doctor of Information Technology.

EDUCATION & CERTIFICATIONS

Trine University, Doctor of Information Technology (DIT) — June 2026 – Present
Wilmington University, Delaware, Master of Science, Information Systems Technologies — Jan 2016 – Dec 2018
Certification: Microsoft Azure Fundamentals (AZ-900)

TECHNICAL SKILLS

Frontend: Angular (2+), TypeScript, RxJS, NgRx, React, Next.js (App Router, SSR/SSG/ISR), JavaScript (ES6+), AngularJS, HTML5, CSS3, SCSS
Architecture & APIs: Frontend Architecture, Design Systems, Component Libraries, Schema-Driven UI, REST APIs, Node.js, Webpack, JWT/RBAC, Stripe
Testing: Jest, React Testing Library, End-to-End Testing, Test-Driven Development (TDD), Jasmine, Karma, SIT, UAT
Cloud & DevOps: Microsoft Azure, Railway, Docker, CI/CD (Jenkins), Git, GitHub, Bitbucket
Databases: PostgreSQL, MongoDB
Practices: Accessibility (WCAG, ARIA), Performance Optimization, AI-Assisted Development, Code Review, Agile, Jira, Confluence

INDEPENDENT PROJECT

AI-Powered Supply Chain Forecasting Platform (CenVexa) — Jan 2026 – Present
Next.js, React, TypeScript, PostgreSQL, Stripe, Railway

- Designed and built a multi-tenant SaaS platform end-to-end, from frontend architecture and REST APIs to database and deployment
- Built a reusable UI component set and real-time visualization dashboards that present demand forecasts across 1,000+ SKUs
- Built a schema-driven UI layer that renders dashboards and components from JSON configuration instead of hard-coded views
- Implemented JWT authentication with role-based access control (RBAC) across 3 subscription tiers, plus Stripe billing
- Integrated an AI demand forecasting engine (85%+ accuracy) with the frontend to support data-driven planning workflows
- Built a conversational AI assistant that lets non-technical users query complex supply chain data in natural language, with the model's response selecting which UI component renders at runtime
- Practiced test-driven development with Jest unit tests and end-to-end tests that run in the CI/CD pipeline before each deployment
- Built and published CenVexa Kids, an educational mobile app for young children, on the Apple App Store; Android release in preparation

PROFESSIONAL EXPERIENCE

Medline Industries, Chicago, IL — Feb 2022 – Jan 2026
Developer Analyst

- Built an enterprise React + SCSS UI platform with the UI team, adopted across 7+ applications and multiple product teams, improving design consistency and developer productivity
- Led the adoption of the UI Kit across product teams, onboarding engineers, and guiding their migration to shared components and patterns
- Mentored developers on React, TypeScript, and accessibility practices, and owned code reviews for UI Kit contributions to keep quality and consistency high
- Developed and maintained components in Medline's UI Kit libraries (UI Line and React components), improving modularity, reuse, and maintainability
- Delivered accessible, reusable components that meet WCAG and ARIA standards, with full keyboard navigation
- Standardized frontend architecture and UI patterns in partnership with engineers across teams, reducing duplicated UI code
- Built user-facing features in React and TypeScript, covered by Jest and React Testing Library tests that ran in CI/CD pipelines for reliable delivery

Bank of America, Jersey City, NJ — June 2020 – Dec 2021
UI Developer

- Developed UI components in AngularJS, JavaScript, HTML5, and CSS3 in collaboration with cross-functional teams
- Carried out SIT and UAT testing to verify stability and reliability before release

Dun & Bradstreet, Center Valley, PA (iresearch.dnb.com) — Jan 2019 – June 2020
UI Developer

- Migrated a legacy AngularJS application to Angular (2+) and TypeScript as part of the UI team, improving structure, performance, and maintainability
- Built reusable components, services, and feature modules with Angular CLI, shared across the application
- Managed application state with NgRx and RxJS Observables, and integrated REST APIs through Angular services
- Implemented Reactive Forms with validation and Angular Router navigation, using lazy-loaded modules to improve load performance
- Wrote unit tests for components and services in Jasmine and Karma
- Took part in code reviews and Agile ceremonies, including sprint planning and daily stand-ups

PUBLICATIONS & RESEARCH

- "Developer Productivity and Project Cost Savings Through Component Reusability," *Global Journal of Engineering and Basic Sciences (GBEJ)*, 2025.
- "Designing Resilient Front-End Architectures for Real-Time Web Applications," *International Journal of Engineering Technology Research & Management (IJETRM)*, 2024. DOI: 10.5281/zenodo.15486443
- "Bridging the Gap: Integrating AI-Assisted Code Suggestions in Your UI Workflow," *International Journal of Engineering Technology Research & Management (IJETRM)*, 2023. DOI: 10.5281/zenodo.15321149